

Artista 180 Address Space

The 24-bit address space as this machine actually wires it, read off a full 16 MB memory dump and the on-chip layout in the hardware manual. Only about 4 MB responds; the rest is empty air.

- Flash
- Off-chip SRAM
- On-chip SRAM
- On-chip registers
- Memory-mapped device
- Observed, purpose unestablished

H'000000 H'002FFF	Boot ROM 12K Exception vectors, reset at <code>boot_reset</code> H'000400, and the serial API — <code>read_serial_data_byte</code>, <code>send_serial_data_byte</code>, <code>get_sci_rx_ready_bit</code>. Byte-identical copies appear at H'008000, H'010000 and H'018000: the device decodes only the low 15 bits, so it repeats every 32K.
H'020000 H'02000F	SED1351F display registers 15 bytes The LCD controller's register window, on its own chip select at the foot of the RAM bank. <code>lcd_controller_init</code> H'20E062 writes H'020001–H'02000F once each, then enables the panel by writing H'DB back to the first. Never read anywhere in the image — write-only. H'020002 = H'4F (79, one less than the 80 bytes per line) and H'020004 = H'EF (239, one less than 240 lines) match the frame buffer exactly.
H'020010 H'03FFFF	External RAM — working storage 128K SCIO receive state and payload buffer live here: <code>sci0_rx_payload</code> H'0203D4, <code>sci0_frame_len</code> H'020886, <code>sci0_rx_state</code> H'02089E.
H'040000 H'044AFF	LCD frame buffer 18.75K 320 × 240 at 2 bits per pixel, MSB first — 19,200 bytes. Behind the Epson SED1351F controller and its two 32K × 8 SRAMs. The upper half of the same bank, from H'048000, holds dense non-display data rather than a second frame buffer.
H'050000 H'05FFFF	Alias of bank 04 64K The same SRAM seen twice: A16 is not decoded, so bank 04 reappears at bank 05, byte for byte across all 64K. Anything written to the frame buffer at H'040000 can be read back at H'050000. A dump therefore captures the screen image twice, which is worth knowing — in the simulator the copy at H'050000 goes stale, because the loader treats the two banks as separate memory and only H'040000 gets redrawn.
H'060000 H'07FFFF	Reads H'FF everywhere 128K Nothing drives the bus here — an idle, pulled-up window.

H'080000 H'09FFFF	Digital input latch 128K window One byte, mirrored at every address in the window because the board ignores the low address lines. <code>input_scan</code> H'209E48 polls it; bits 1 and 2 feed <code>mode_FFEC0</code> . Reads H'70 idle.
H'0A0000	Digital output latch write-only Written by the port-setup code at H'20BFC4 onward; reads back as nothing, so it never appears in the dump.
H'0E0000 H'0ECFFF	External RAM — cleared at boot 44K Bulk-written at boot through the copy routine at H'2003A0 — 512 writes per page across the whole span, so this is working memory rather than a device.
H'114000 H'11BFFF	External RAM — application state 32K Where the machine keeps its live state: <code>adc_results</code> H'11A251, the touch debounce counters, the calibration constants copied out of flash at H'11A87E, and the touch position in pixels at <code>touch_x_pixels</code> H'11B102.
H'11F000 H'11FFFF	External RAM — hot page 4K The most heavily written page in the machine: 15.9 million writes in 40 million instructions.
H'200000 H'3FFFFFF	Application flash 2M The interrupt handler table sits at H'200000 — the boot ROM's trampolines fetch real handlers from it — followed by the firmware itself, roughly 71,500 instructions reached by static analysis, and the icon bitmaps. Version string <code>NMMV03.01</code> at H'200100.
H'500000 H'5FFFFFF	Data flash 1M Programmed in place — the firmware issues the AMD unlock sequence to H'575555 and H'572AAA. Holds the configuration block at H'57FF80: the H'A5 validity marker, the four touch-calibration floats at H'57FFA0, and the version string.
H'FFF710 H'FFFFFF	On-chip RAM and registers 2K + 228B Detailed below.

THE ON-CHIP PAGE, H'FFF710 – H'FFFFFF

H'FFF710 – H'FFFF0F	On-chip SRAM, 2K. The stack starts at H'FFFD4 and grows down — <code>boot_reset</code> 's first instruction sets it. Machine state lives at the top: <code>CHAN_SELECTION</code> H'FFFD1C, <code>input_flags</code> H'FFFE1, and the touch coordinates <code>touch_x_raw</code> H'FFFD9 / <code>touch_y_raw</code> H'FFFEA that the calibration turns into pixels.	2,048 B
H'FFFF10 – H'FFFF1B	Reserved.	12 B

H'FFFF1C – H'FFFFFF

On-chip registers: DMAC H'FFFF20, ITU H'FFFF60, ports H'FFFFC5, SCIO and SCI1 H'FFFFB0, A/D H'FFFFE0, bus controller H'FFFFEC.

228 B

SUMMARY

RANGE	SIZE	KIND	BASIS
H'000000–H'002FFF	12K	Boot ROM	ESTABLISHED
H'020000–H'02000F	15B	SED1351F display registers	ESTABLISHED
H'020010–H'044FFF	148K	Off-chip SRAM, incl. frame buffer	ESTABLISHED
H'050000–H'05FFFF	64K	Alias of bank 04 (A16 undecoded)	ESTABLISHED
H'060000–H'07FFFF	128K	Idle bus	OBSERVED
H'080000–H'09FFFF	128K	Digital input latch	ESTABLISHED
H'0A0000	–	Digital output latch	ESTABLISHED
H'0E0000–H'0ECFFF	44K	Off-chip SRAM, cleared at boot	ESTABLISHED
H'114000–H'11FFFF	36K	Off-chip SRAM, application state	ESTABLISHED
H'200000–H'3FFFFFF	2M	Application flash	ESTABLISHED
H'500000–H'5FFFFFF	1M	Data flash	ESTABLISHED
H'FFF710–H'FFFF0F	2K	On-chip SRAM	ESTABLISHED
H'FFFF1C–H'FFFFFF	228B	On-chip registers	ESTABLISHED

Regions come from scanning Bernina180_20260816.bin at 4K granularity and from watching the firmware run on the simulator; the on-chip boundaries are from the H8/3003 hardware manual. The vertical scale is broken — a linear one would render every mapped region invisible against 16 MB of empty space. Which physical chip covers which range is inferred from where the bus responds, not from the schematic.